LECTURE-6

Cell

Acknowledgements: Some of the materials are taken from v different sources in Google besides from Campbell's book on biology

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Overview

KEY CONCEPTS

6.1 Biologists use microscopes and the tools of biochemistry to study cells

6.2 Eukaryotic cells have internal membranes that compartmentalize their functions

6.3 The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes

6.4 The endomembrane system regulates protein traffic and performs metabolic functions in the cell

6.5 Mitochondria and chloroplasts change energy from one form to another

6.6 The cytoskeleton is a network of fibers that organizes structures and activities in the cell

6.7 Extracellular components and connections between cells help coordinate cellular activities

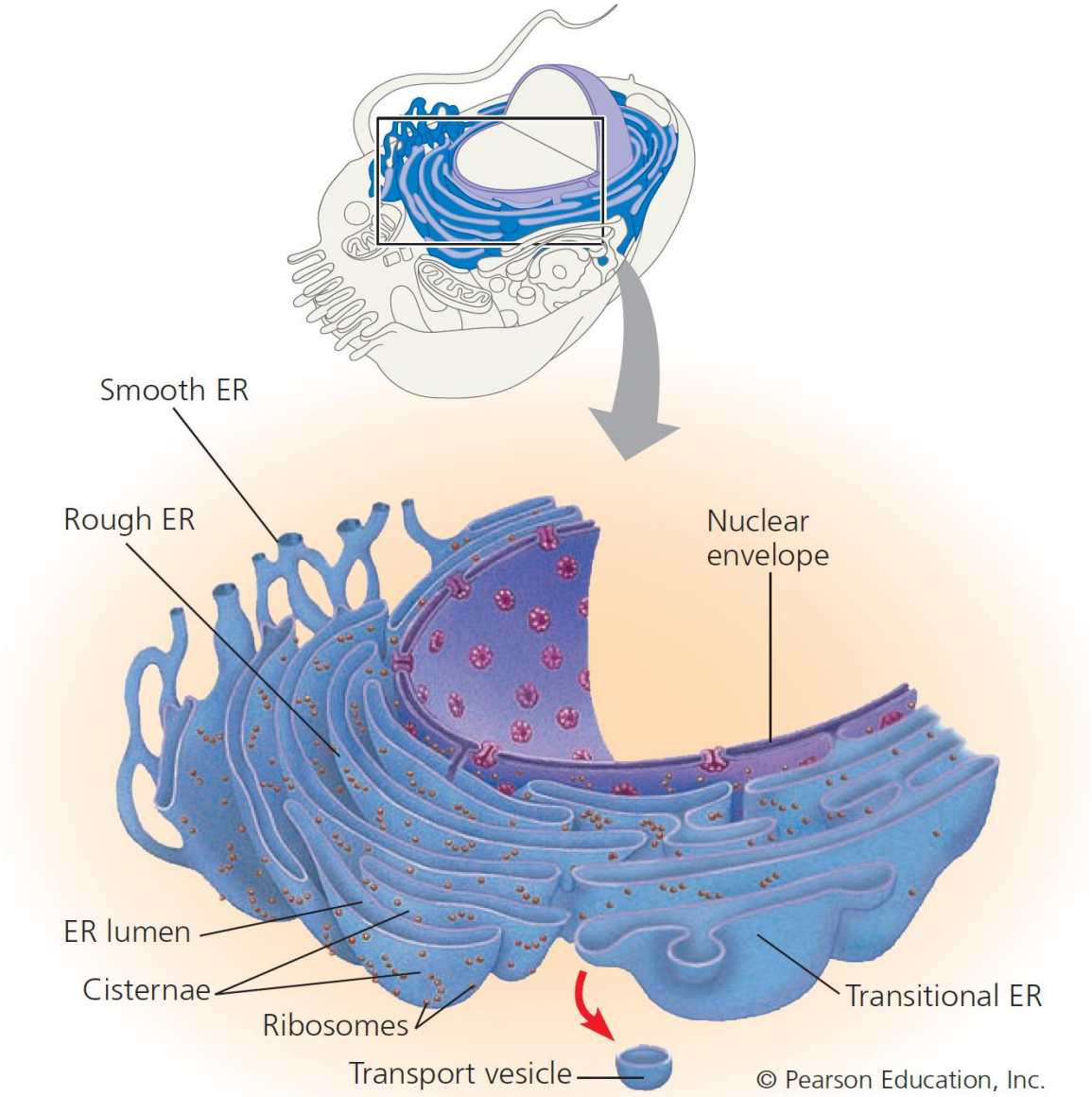
Endomembrane system

- Function: The endomembrane system regulates protein traffic and performs metabolic functions in the cell

- What are the endomembrane systems?

Many of the internal membranes in a eukaryotic cell are part of the **endomembrane system**, which includes the

- ***nuclear envelope,***
- ***endoplasmic reticulum***
- ***Golgi apparatus***
- ***lysosomes,***
- ***vesicles, vacuoles, and plasma membrane.***

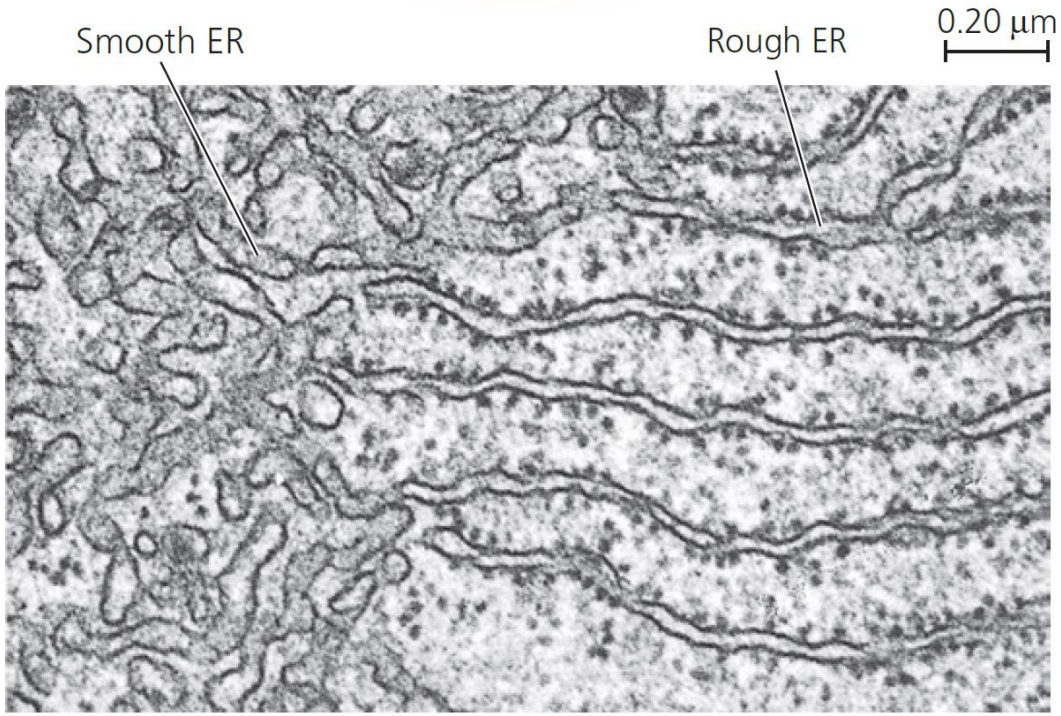


Endomembrane system

- The tasks of the endomembrane system include **synthesis of proteins and their transport into membranes and organelles or out of the cell**, metabolism and movement of lipids, and detoxification of poisons.

- These membranes are either directly continuous or connected via the transfer of **vesicles**, sacs of membrane.

- In spite of the connections, these *membranes are diverse in function and structure*.
  - *The thickness, molecular composition, and types of chemical reactions carried out by proteins in a given membrane may be modified several times during a membrane's life*.



▲ **Figure 6.11 Endoplasmic reticulum (ER).** A membranous system of interconnected tubules and flattened sacs called cisternae, the ER is also continuous with the nuclear envelope, as shown in the cutaway diagram at the top. The membrane of the ER encloses a continuous compartment called the ER lumen (or cisternal space). Rough ER, which is studded on its outer surface with ribosomes, can be distinguished from smooth ER in the electron micrograph (TEM). Transport vesicles bud off from a region of the rough ER called transitional ER and travel to the Golgi apparatus and other destinations.

Endoplasmic reticulum

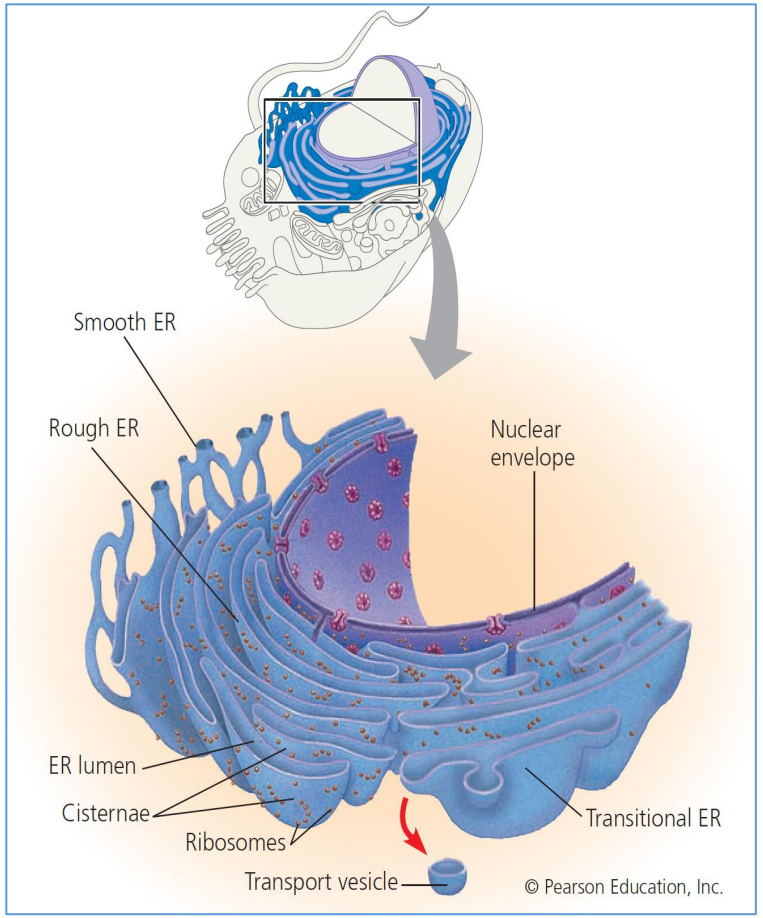
The endoplasmic reticulum manufactures membranes and performs many other biosynthetic functions.

- The **endoplasmic reticulum (ER)** accounts for more than half the membranes in a eukaryotic cell.

- The ER includes a network of membranous tubules and sacs called cisternae that separate the internal compartment of the ER, the ER lumen or cisternal space, from the cytosol.

- The ER membrane is continuous with the nuclear envelope, and the cisternal space of the ER is continuous with the space between the two membranes of the nuclear envelope.

- There are two connected regions of ER that differ in structure and function.
  - ✓ ***Smooth ER*** *looks smooth because it lacks ribosomes.*
  - ✓ ***Rough ER*** *looks rough because ribosomes are attached to the outside, including the outside of the nuclear envelope.*



Role of smooth ERs

- The smooth ER functions in diverse metabolic processes, which vary with cell type. These processes include the

- ✓ synthesis of lipids,
- ✓ metabolism of carbohydrates,
- ✓ detoxifying drugs and poisons, and
- ✓ storing calcium ions.

- Enzymes of the smooth ER are important in synthesising lipids, including oils, steroids, and new membrane phospholipids.

- Other enzymes of the smooth ER help detoxify drugs and poisons, especially in liver cells.

- Detoxification usually involves adding hydroxyl groups to drug molecules, making them more soluble and easier to flush from the body.

- The sedative phenobarbital and other barbiturates are examples of drugs metabolized in this manner by smooth ER in liver cells.

- Barbiturates, alcohol, and many other drugs induce the proliferation of smooth ER and its associated detoxification enzymes, thus increasing the rate of detoxification.

- The smooth ER also stores calcium ions.

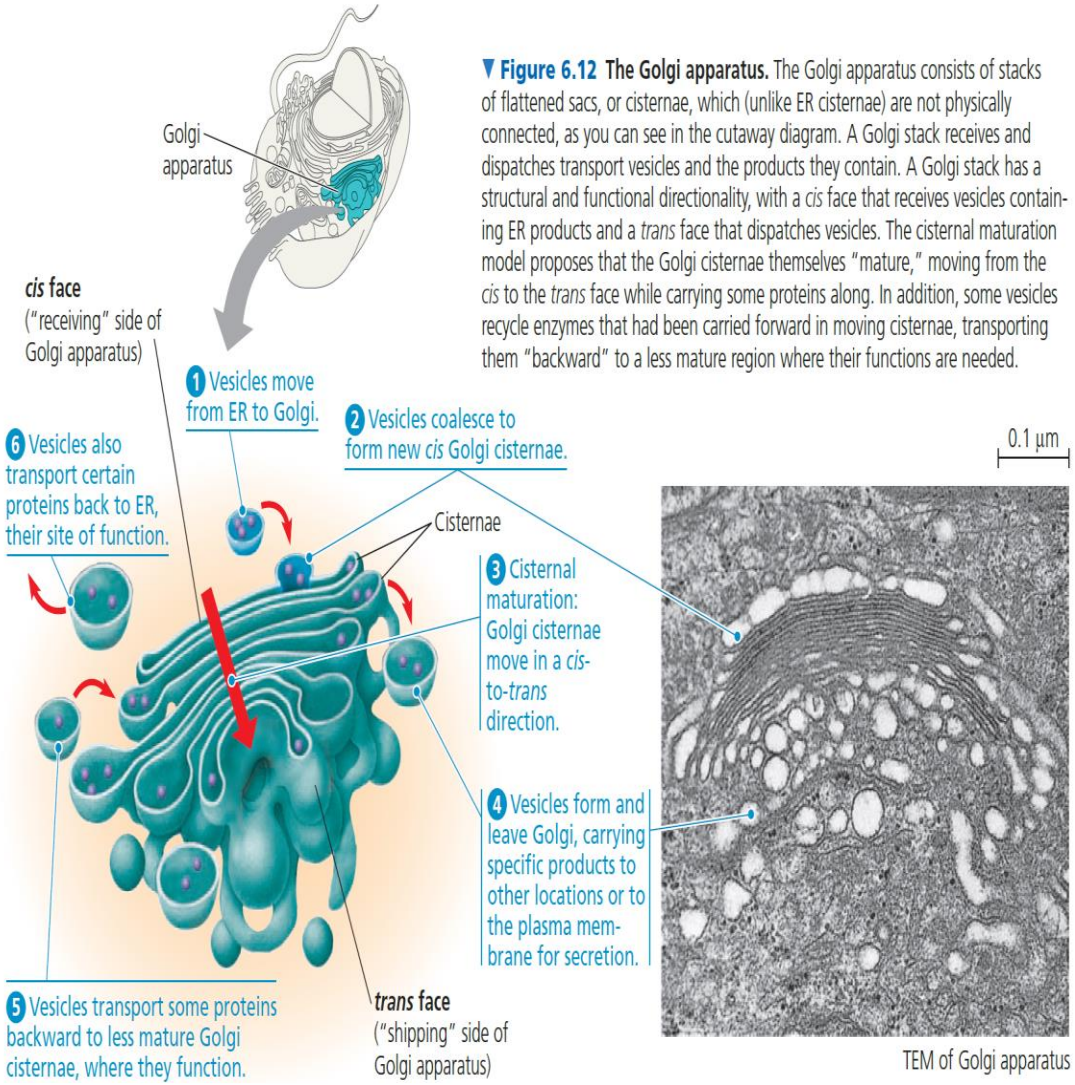
- For example, the smooth ER membrane pumps calcium ions from the cytosol into the ER lumen in muscle cells.

- When a nerve impulse stimulates a muscle cell, calcium ions rush back across the ER membrane into the cytosol and trigger the contraction of the muscle cell.

Role of rough ERs

- Many cells secrete proteins produced by ribosomes attached to rough ER. For example, certain pancreatic cells synthesize the protein insulin in the ER and secrete this hormone into the bloodstream.
- Secretory proteins depart from the ER wrapped in the membranes of vesicles that bud-like bubbles from a specialized region called transitional ER.
- Vesicles in transit from one part of the cell to another are called **transport vesicles**;
- In addition to making secretory proteins, rough ER is a membrane factory for the cell; it grows by adding membrane proteins and phospholipids to its membrane.
- As polypeptides destined to be membrane proteins grow from the ribosomes, they are inserted into the ER membrane and anchored there by their hydrophobic portions.
- Like the smooth ER, the rough ER also makes membrane phospholipids; enzymes built into the ER membrane assemble phospholipids from precursors in the cytosol.
- The ER membrane expands, and portions are transferred to other endomembrane system components as transport vesicles.

Golgi apparatus



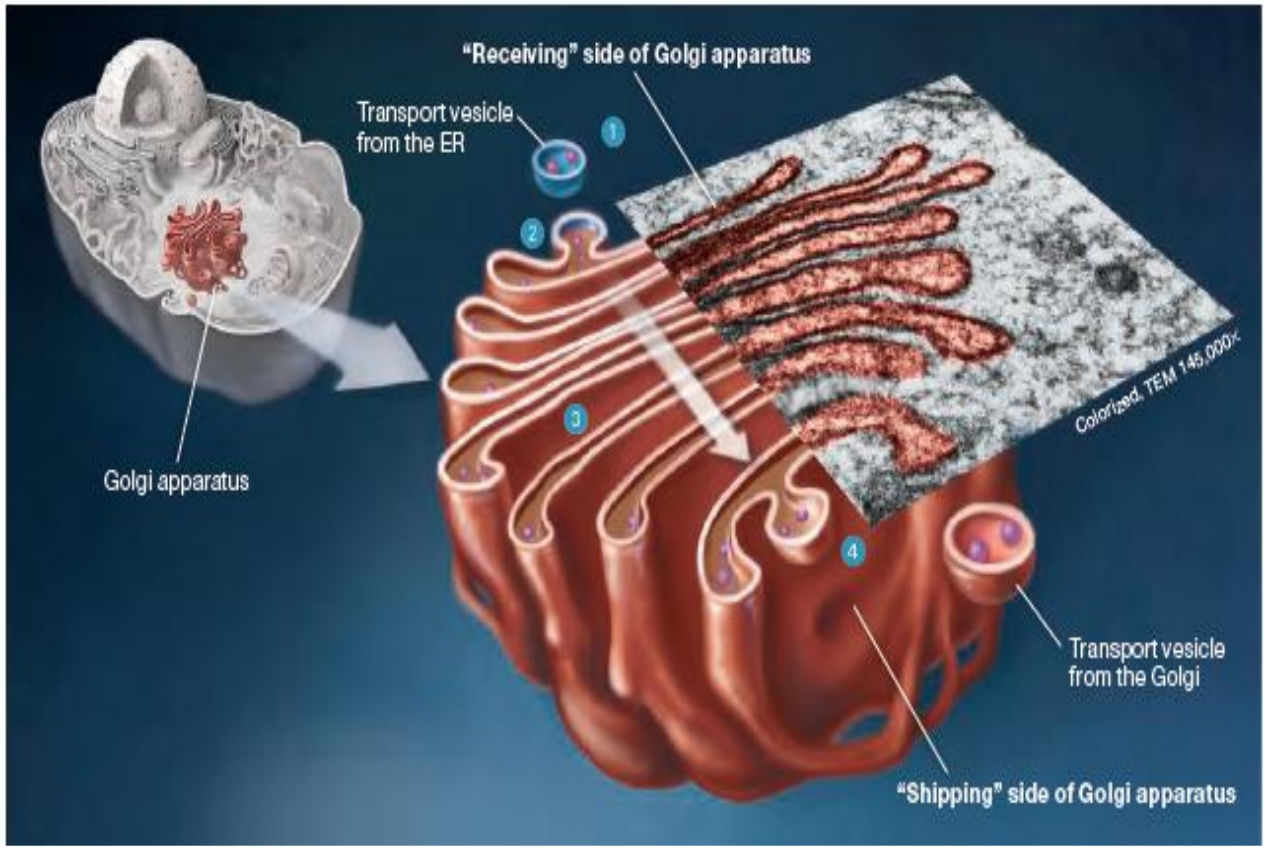
▼ **Figure 6.12 The Golgi apparatus.** The Golgi apparatus consists of stacks of flattened sacs, or cisternae, which (unlike ER cisternae) are not physically connected, as you can see in the cutaway diagram. A Golgi stack receives and dispatches transport vesicles and the products they contain. A Golgi stack has a structural and functional directionality, with a *cis* face that receives vesicles containing ER products and a *trans* face that dispatches vesicles. The cisternal maturation model proposes that the Golgi cisternae themselves "mature," moving from the *cis* to the *trans* face while carrying some proteins along. In addition, some vesicles recycle enzymes that had been carried forward in moving cisternae, transporting them "backward" to a less mature region where their functions are needed.

- ***The Golgi apparatus is the shipping and receiving centre for cell products.***
- Many transport vesicles from the ER travel to the **Golgi apparatus** to modify their contents.
- The Golgi apparatus **is a manufacturing, warehousing, sorting, and shipping centre**.

Golgi-structure: Cis and trans face

- Some products of the ER are modified and stored and then sent on.
- The Golgi apparatus is especially extensive in cells specialized for secretion.
- The Golgi apparatus **consists of flattened membranous sacs**—cisternae—resembling a stack of pita bread.
  - ✓ The membrane of each cisterna separates its internal space from the cytosol.
- One side of the Golgi apparatus, **the *cis* face**, is **near the ER**.
- ✓ The *cis* face **receives material from the ER by fusing with transport vesicles**.

- Products from the ER are usually modified during transit from the *cis* to the *trans* region of the Golgi apparatus.

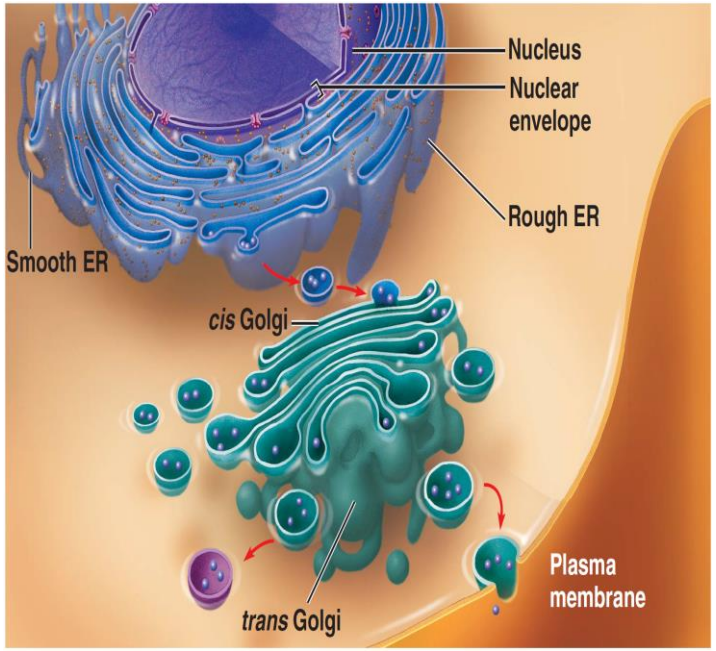


Golgi function: Cisternal maturation model

- ***The Golgi apparatus is the shipping and receiving centre for cell products.***

  - ✓ The Golgi apparatus is **a dynamic structure**.
  - ✓ The Golgi **manufactures and refines its products in stages**, with different cisternae between the *cis* and *trans* regions containing unique teams of enzymes.

- According to the ***cisternal maturation model***, the cisternae of the Golgi progress from the *cis* to the *trans* face, carrying and modifying their protein cargo as they move.



  - ✓ Finally, the **Golgi sorts and packages materials** into transport vesicles.
  - ✓ Molecular **identification tags such as phosphate groups** are added to products to aid sorting.
  - ✓ These **tags act like ZIP codes** on mailing labels to identify the product's final destination.

- Transport vesicles budded from the Golgi may have **external molecules** on their membranes that **recognize "docking sites"** on the surface of specific organelles or the plasma membrane, thus targeting them appropriately.

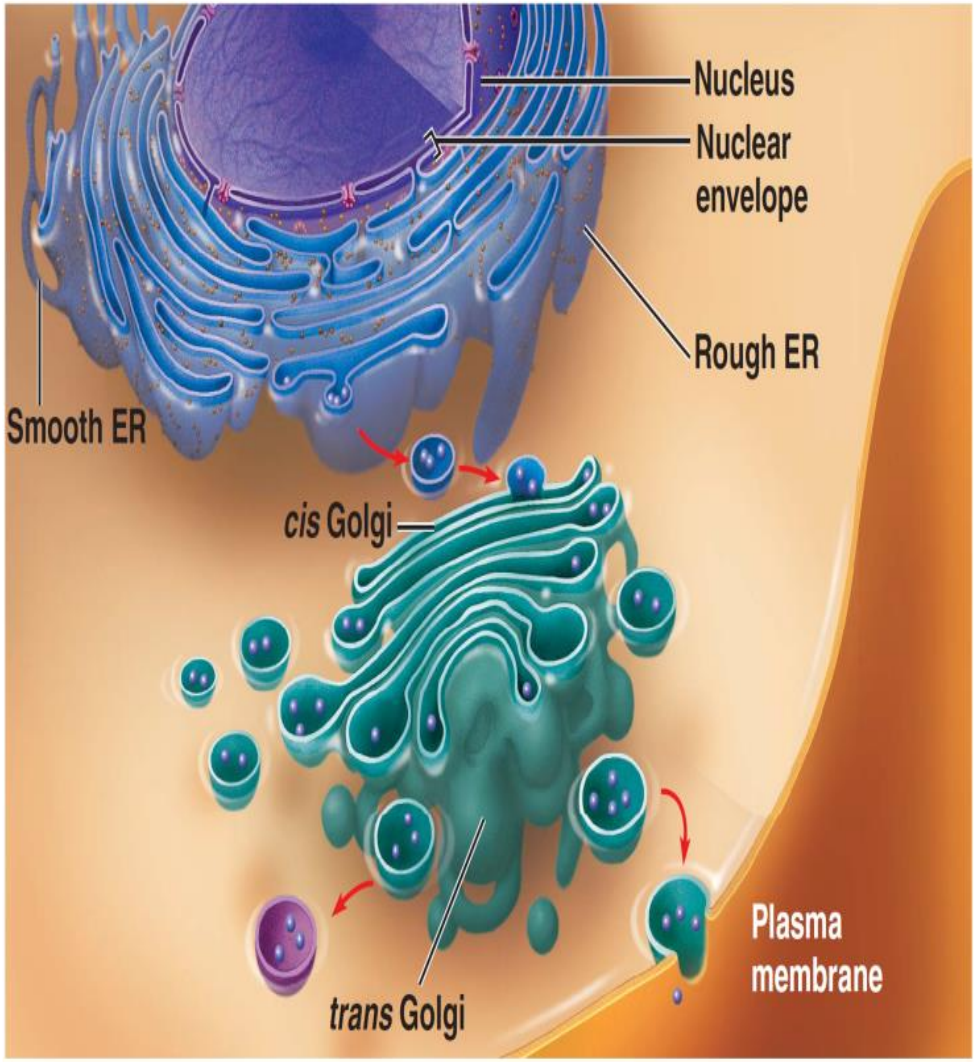
Golgi apparatus: modification of carbohydrates

- ***The Golgi apparatus is the shipping and receiving centre for cell products.***

- Various **Golgi enzymes modify the carbohydrate portions of glycoproteins**.
  - ✓ Carbohydrates are first added to proteins in rough ER, often during polypeptide synthesis.
  - ✓ The carbohydrate on the resulting glycoprotein is modified as it passes through the rest of the ER and the Golgi.
  - ✓ The Golgi removes some sugar monomers and substitutes others, producing many carbohydrates.

- The Golgi can **manufacture its own macromolecules**, including pectin and other non-cellulose polysaccharides.

- Golgi **products that will be secreted depart from the *trans* face** of the Golgi inside transport vesicles that eventually fuse with the plasma membrane.



Vesicles

- Vesicle Functions: Anatomically, a cell comprises various organelles that function in an organized manner to carry out metabolic processes.

- Vesicles are vital because they have a wide variety of functions that contribute to the proper functioning of the cell such as packaging, storage, digestion, transport, cell communication, metabolic pathways and others. Among these, its most important function is that of transport.

- Among these organelles, is a tiny intra or extracellular structure that is enclosed by a lipid membrane, usually carrying fluid; called Vesicle.

- Vesicles are typically transient structures formed during the secretion or uptake of molecules from or into the cell, respectively. Vesicles help in transporting substances in the cell.

Vesicles

- Vesicles are formed when by the pinching off of the cell membrane of the endoplasmic reticulum or Golgi apparatus, or if an extracellular substance gets surrounded by the cell membrane.

- The formation of vesicles involves a set of coat proteins that form the rounded shape of the vesicle. These proteins help to engulf material that needs to be transported, within the vesicle.

- Vesicles can fuse with the cell membrane as well as organelle membranes because they are enclosed by a lipid bilayer. Due to this, they can move in and out of the cell, as well as between organelles like Golgi bodies and Endoplasmic reticulum.

Types of Vesicles

Vacuoles: These are tiny lipid enclosed structures that usually contain water, and are mostly seen in plants and certain bacteria. They are used for regulating osmotic pressure in the cell.

Lysosomes: Lysosomes are a type of vesicle which is involved in cellular digestion. A lysosome contains proteolytic enzymes that can break down food molecules.

Peroxisomes: Similar to lysosomes, peroxisomes are specialized vesicles that contain hydrogen peroxide. These vesicles are primarily involved in cellular oxidation reactions.

Transport vesicles: As the name suggests, these are tiny sacs enclosed by a lipid bilayer that is heavily involved in the transport of materials from and to the cell and between organelles.

Secretory vesicles: A type of specialized vesicle that carries substances out of the cell. They usually generate from the Golgi apparatus.

Vesicles

Synaptic vesicles: A type of specialized vesicle found in the types of neurons that store and transport neurotransmitter molecules.

Extracellular vesicles: Extracellular vesicles are found outside the cell and are used for transport into the cell. These type of vesicles are seen in both eukaryotic and prokaryotic cells.

Gas vesicles: These are found in bacteria and provide buoyancy to the cell.

Transport by vesicles: Endo and Exocytosis

- The primary purpose of vesicles is the transport of materials between organelles, and into the cell.

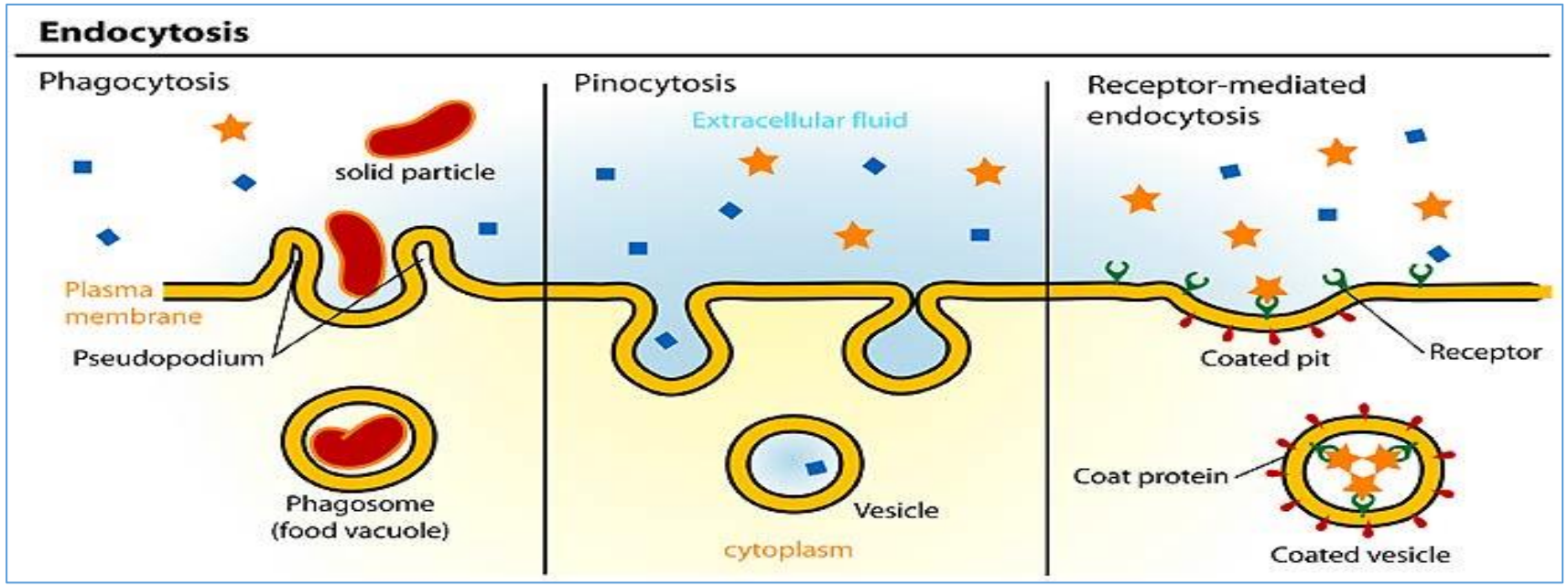
- Different types of transport vesicles are found budding off and transporting substances from the smooth endoplasmic reticulum to the rough endoplasmic reticulum for processing, as well as from the Golgi apparatus.

- Vesicles also help in carrying substances into the nucleus across then nuclear membrane. Proteins that require processing are transported throughout the cell from one organelle to another in vesicles.

- There are two major types of transport mechanisms that involve the use of vesicles.

  - ✓ Endocytosis
  - ✓ Exocytosis

Exocytosis and Endocytosis



- Endocytosis is a process whereby substances and molecules are transported into the cell from the extracellular environment. This process uses vesicles as the primary means of transport.

- There are three types of endocytosis –
  - ✓ Phagocytosis,
  - ✓ Pinocytosis, and
  - ✓ Receptor-mediated phagocytosis.

Endocytosis

- **Phagocytosis** involves the formation of a food vesicle following engulfment of food particles or whole cells like bacterial cells. This results in the formation of a vesicle known as the phagosome.

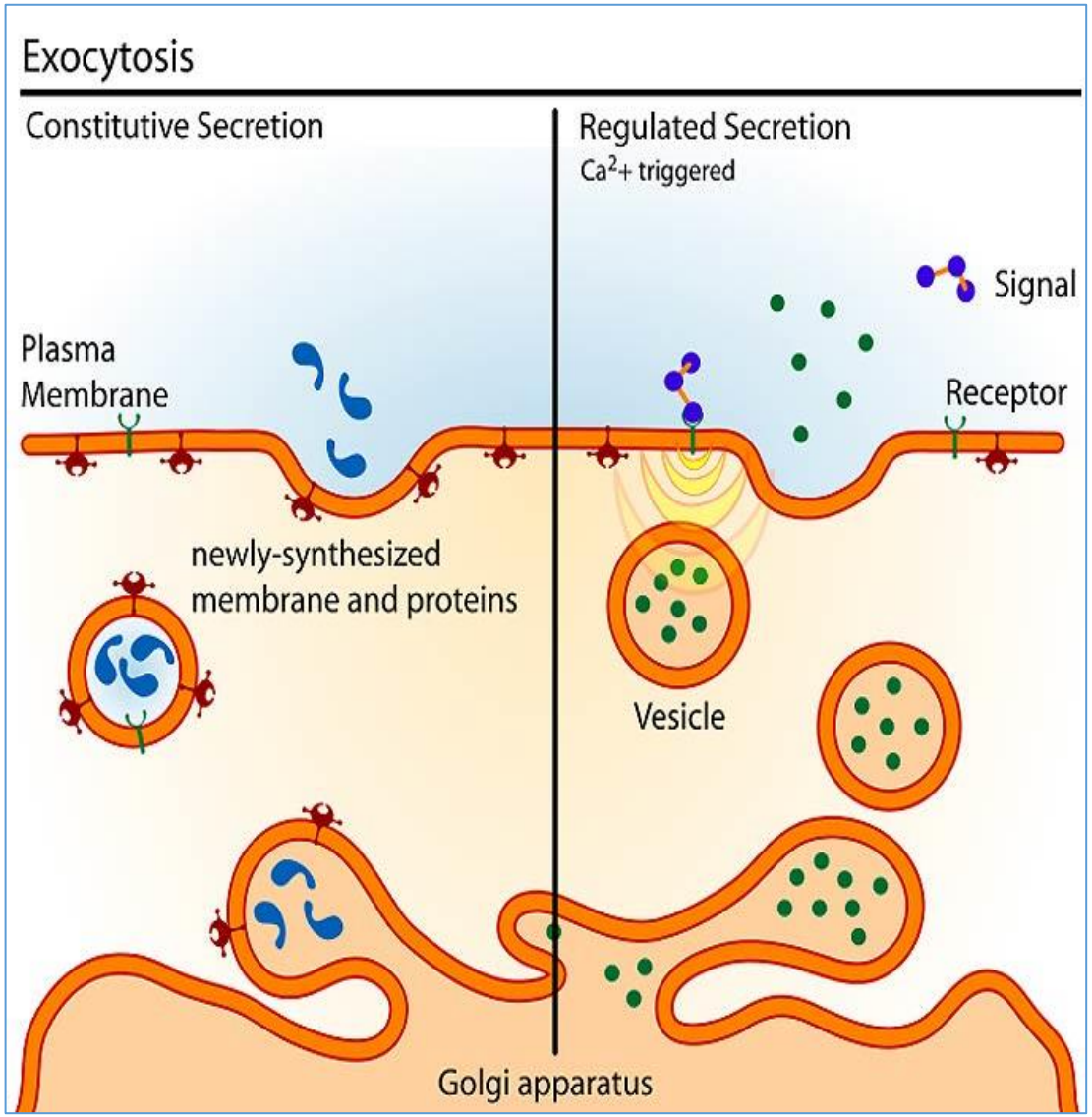
- The phagocytic vesicle then fuses with a lysosome to digest and break down the contents.

- **Pinocytosis** is a similar process, but it includes fluid molecules or water being engulfed by tiny vesicles into the cell.

- **Receptor-mediated endocytosis** is a type of endocytosis that allows uptake of particular molecules into the cell based on the presence of receptors specific to that molecule on the cell membrane.

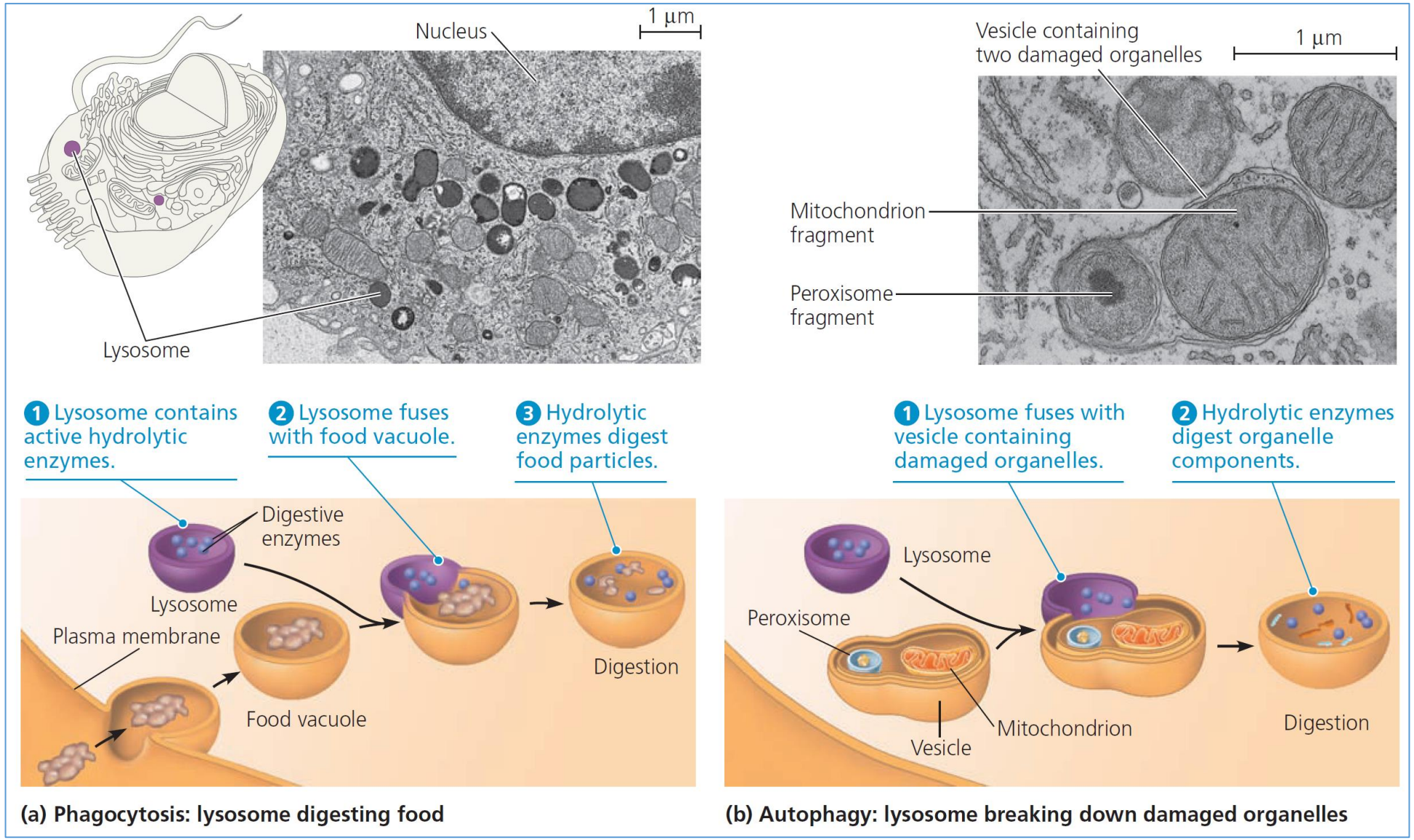
- Once a molecule has been selected, it is transported in vesicles.

Exocytosis



- Exocytosis is a process where substances move out of the cell.
- Secretory vesicles that bud off the Golgi apparatus travel to the cell membrane where they fuse and release contents to the exterior of the cell.

Lysosomes- digestive compartments



(a) Phagocytosis: lysosome digesting food

(b) Autophagy: lysosome breaking down damaged organelles

Lysosomes are digestive compartments.

- A lysosome is a membrane-bound sac of hydrolytic enzymes that an animal cell uses to digest macromolecules.
- Lysosomal enzymes work best at acidic pH.
  - ✓ *The rupture of one or a few lysosomes has little impact on a cell because the lysosomal enzymes are not very active at the neutral pH of the cytosol.*
  - ✓ *Massive rupture of many lysosomes can destroy a cell by **autodigestion**.*

- Lysosomal enzymes and membranes are synthesized by rough ER and then transferred to the Golgi apparatus for further modification.
  - ✓ *Some lysosomes arise by budding from the trans face of the Golgi apparatus.*

- **Proteins on the inner surface of the lysosomal membrane are spared** by digestion by their three-dimensional conformations, which protect vulnerable bonds from hydrolysis.
- Lysosomes carry out **intracellular digestion** in a variety of circumstances.
  - ✓ *Amoebas eat by engulfing smaller organisms by **phagocytosis**.*
  - ✓ *The food vacuole formed by phagocytosis fuses with a lysosome, whose enzymes digest the food.*
  - ✓ *As the polymers are digested, monomers pass to the cytosol to become nutrients for the cell.*
  - ✓ *Human macrophages help defend the body by phagocytosing and destroying invasive bacteria.*

Lysosomes- role

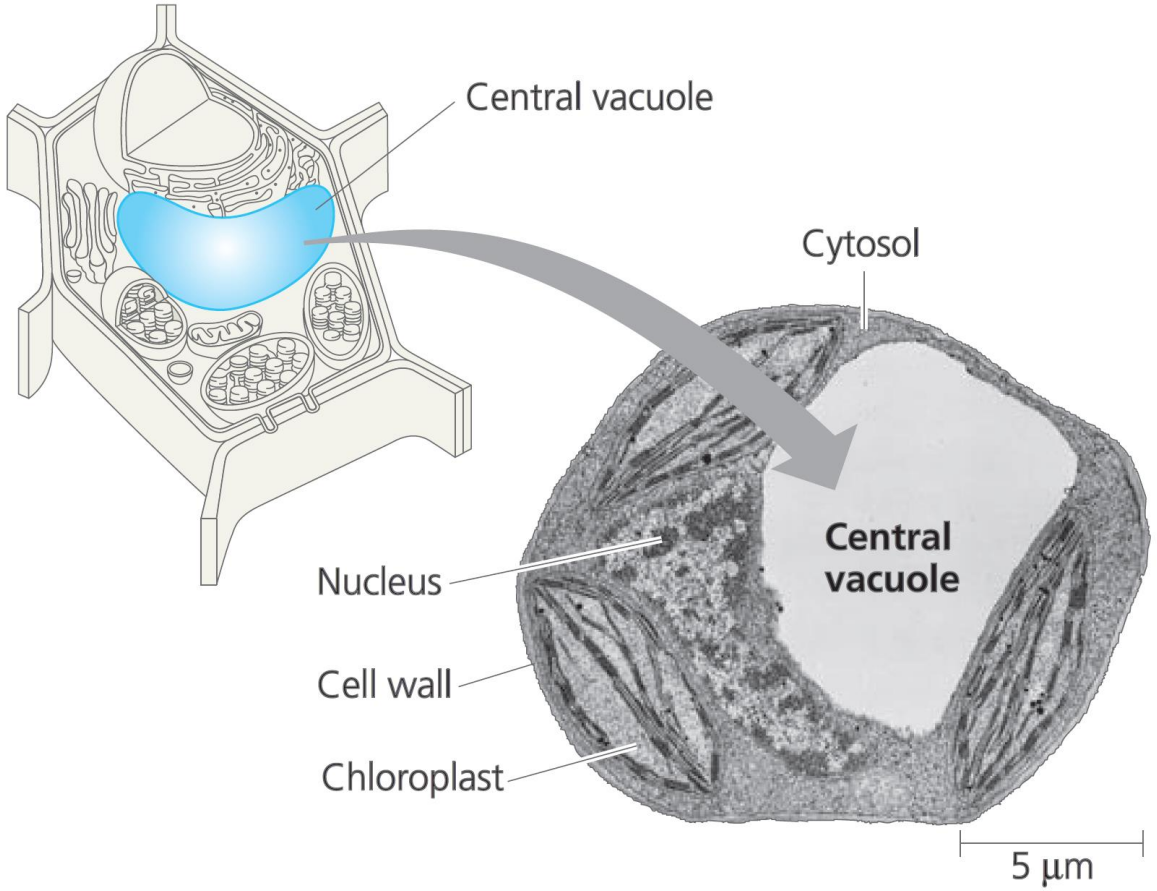
- Lysosomes can play a role in **recycling the cell's organelles and macromolecules**.
- This **recycling**, or ***autophagy***, renews the cell.

  - ✓ *During autophagy, a damaged organelle or region of cytosol becomes surrounded by a double membrane of unknown origin.*
  - ✓ *A lysosome fuses with the outer membrane of the vesicle, digesting the macromolecules and returning the organic monomers to the cytosol for reuse.*

- The cells of people who have inherited **lysosomal storage diseases** lack a functioning hydrolytic enzyme normally present in lysosomes.

  - ✓ *The **lysosomes become engorged with indigestible substrates**, which begin to interfere with other cellular activities.*
  - ✓ *In people who have **Tay-Sachs disease**, a lipid-digesting enzyme is missing or inactive, and the brain becomes impaired by an accumulation of lipids in the cells.*

Vacuole – structure



▲ **Figure 6.14 The plant cell vacuole.** The central vacuole is usually the largest compartment in a plant cell; the rest of the cytoplasm is often confined to a narrow zone between the vacuolar membrane and the plasma membrane (TEM).

Vacuole- function

Vacuoles have diverse functions in <u>cell maintenance</u>.

- Vacuoles are **large vesicles derived from the EG and Golgi apparatus**.
- These membrane-bound sacs have a variety of functions.
- **Food vacuoles** are formed by phagocytosis and fuse with lysosomes.
- **Contractile vacuoles**, found in freshwater protists, pump excess water out of the cell to maintain the appropriate concentration of ions and molecules inside the cell.

Vacuole- function

- **In plants and fungi, vacuoles carry out enzymatic hydrolysis**, like animal lysosomes do.

- A large **central vacuole** is found in many mature plant cells.
  - ✓ *The vacuolar membrane is selective in its transport of solutes into the central vacuole.*
  - ✓ *As a result, the solution inside the vacuole, called cell sap, differs in composition from the cytosol.*

- The functions of the central vacuole include **stockpiling proteins or inorganic ions, disposing of metabolic by-products, holding pigments, and storing defensive compounds** that *protect the plant against herbivores*.
  - The vacuole has a major role in the growth of plant cells, which enlarge as their vacuoles absorb water, enabling the cell to become larger with little investment in new cytoplasm.

- Because of the large vacuole, the cytosol occupies only a thin layer between the plasma membrane and the central vacuole. The presence of a large vacuole increases the ratio of surface area to volume for the cell.